

decomposed, it results in the following nonlinear form:

$$\begin{aligned} \ln(\text{WTP}) = & \beta \ln(\eta_1 \text{QWBSYMSCORE} \\ & + \eta_2 \text{QWBMOBScore} \\ & + \eta_3 \text{QWBSACScore} \\ & + \eta_4 \text{QWBPAcScore}) \\ & + \beta \ln(\Delta \text{DAYS}/365) + \alpha X \end{aligned} \quad (9)$$

We estimated the nonlinear models with the QWB subcomponent scores using maximum likelihood estimation and report the results in Table 5. In each of these specifications, one of the η coefficients (in this case η_4) is restricted to be equal to one. This restriction implies that, if the marginal effects of the four scores on WTP are the same, then the other three η coefficients will also equal one. Furthermore, if these coefficients are all equal to one, then the specification in Equation (9) simplifies to Equation (8).

The results of these nonlinear specifications again cast doubt on the assumption underlying the composite QWB score. The coefficients on the mobility and physical activity scores are distinctly larger than for the other two QWB scores. To test the restrictions implied by using the sum of the four scores (ΔQWB) as a composite index, in these cases we can compare the log-likelihood values for the restricted models (in Table 4) with those for the unrestricted models (in Table 5). Based on likelihood ratio tests, the restricted models can all be rejected at a 0.05 level of significance.

In the models reported in Table 5, the effects of duration income, age, and other variables are, for the most part, consistent with the previous results. The coefficients on ΔDAYS (linear and log form) are virtually unchanged and consistently positive and significant. The effect of $\ln(\text{INCOME})$ is also consistently positive and significant, although the coefficient is somewhat larger. The final specification in Table 5 implies an income elasticity of almost 0.9. The estimated elasticity of WTP with respect to age continues to be positive and significant, and once again it is surprisingly large (i.e. greater than 3 in Table 5). The effect of gender composition (%MALE) continues to be negative but is somewhat larger (in absolute value) and more significant.

Testing the QALY valuation assumptions. The results of the log-linear specifications in both Tables 4 and 5 can also be used to test the assumptions of

the QALY valuation approach for estimating morbidity values. The results do not support these assumptions.

According to the QALY valuation approach, WTP is assumed to increase in direct proportion to the gain in QALYs. As previously discussed, it is often assumed that the value per QALY is approximately \$100 000. Under less restrictive assumptions, the relationship between WTP and QALYs can be expressed as

$$\text{WTP} = \alpha(\Delta \text{QALY})^\beta \quad (6')$$

In other words, the QALY valuation approach restricts β to be equal to one. If β is greater than or equal to one, then WTP will increase more or less than proportionately with respect to QALY gains.

If we assume that the QWB Scale is an appropriate health utility index, and we assume that the QALY gain of interest is brought about by avoiding a health effect of specific duration and severity, then Equation (6') can be expanded as follows:

$$\text{WTP} = \alpha(\Delta \text{DAYS}/365 \times \Delta \text{QWB})^\beta \quad (10)$$

In this expression, the QALY gain is expressed as the constant per-period utility gain (from less than 1 to 1) times the duration of the utility gain (converted to years). In log-linear form, this equation becomes

$$\begin{aligned} \ln(\text{WTP}) = & \ln(\alpha) + \beta \ln(1/365) \\ & + \beta \ln(\Delta \text{DAYS}) + \beta \ln(\Delta \text{QWB}) \end{aligned} \quad (11)$$

This expression implies that, with a log-linear model, the restrictions imposed by the QALY valuation approach can be tested directly. The null hypothesis is that coefficients on $\ln(\Delta \text{DAYS})$ and $\ln(\Delta \text{QWB})$ are both equal to one. An F-test of this restriction for the log-linear specification in Table 4 can be strongly rejected (p -value < 0.001). In the last specification, the coefficient estimate for $\ln(\Delta \text{DAYS})$ is considerably less than one (0.5), and the coefficient estimate for $\ln(\Delta \text{QWB})$ is considerably greater than one (1.97). Therefore, even the less restrictive assumptions implied by Equation (6') – that coefficients on $\ln(\Delta \text{DAYS})$ and $\ln(\Delta \text{QWB})$ are equal to one another (but not necessarily equal to one) – can be strongly rejected (p -value = 0.026).

The results of the log-linear specification indicate that WTP increases less than proportionately with respect to changes in duration, as measured by ΔDAYS . This result is consistent with a declining marginal disutility with respect to duration

of illness. Furthermore, the coefficient for $\ln(\Delta QWB)$ implies that WTP increases more than proportionately with the severity of illness. This result is consistent with a decreasing marginal utility with respect to health status.

The final (log-linear) specification in Table 5 can also be used to test the restrictions implied by the QALY valuation approach. By replacing ΔQWB in Equation (11) with a reweighted composite score based on the results in Table 5 ($\eta_1 QWBSYMSCORE + \eta_2 QWBMOBSCORE + \eta_3 QWBSACSCORE + 1 QWBPACSCORE$), it is again possible to test the restriction that the coefficient for $\ln(DAYS)$ is equal to the β parameter. Using a likelihood ratio test, this restriction can again be rejected at a 0.05 level of significance.

Implications for policy analysis

The overall goodness of fit and statistical significance of the models reported in Tables 4 and 5 suggest that they provide a reasonable foundation for constructing a predictive WTP function. In

other words, the models can be used to develop and test alternative benefit transfer functions.

The advantage of being able to specify a WTP function is that it can be used to extrapolate beyond the existing set of WTP estimates in the literature. That is, it can be used to estimate values for any number of avoided acute illnesses, as long as these illnesses can be described according to their severity (in this case, using the QWB classification system) and their duration. In addition, depending on the model specification and econometric results, it can be used to tailor WTP estimates according to the socio-demographic characteristics of the affected population (e.g. average age, gender composition, and/or average income). Developing value estimates that vary according to income levels can raise difficult ethical issues. Nonetheless, it may be useful, for example, to estimate WTP for a person with average income or to estimate how average WTP will be affected by growth in per-capita income levels.

To demonstrate and evaluate how the results can be used to predict WTP for avoiding acute effects under specific conditions, we selected two specifications – the log-linear specification using the composite QWB score (last specification in Table 4) and the log-linear form using the four

Table 6. Benefit transfer function estimates

Explanatory variable	BT function 1 ^a		Explanatory variable	BT function 2 ^b	
	Coefficient	<i>t</i> -stat ^c		Coefficient	<i>t</i> -stat ^c
$\ln(\Delta DAYS)$	0.501	13.07	$\ln(\Delta DAYS)$	0.477	12.93
$\ln(\Delta QWB)$	2.339	6.23	β^d	1.239	2.22
			$QWBSYMSCORE (\eta_1)^d$	0.357	1.31
			$QWBMOBSCORE (\eta_2)^d$	2.120	4.45
			$QWBSACSCORE (\eta_3)^d$	0.476	2.22
			$QWBPACSCORE (\eta_4)^d$	1.000	
$\ln(INCOME)$	0.777	3.01	$\ln(INCOME)$	0.833	3.96
$\ln(AGE)$	2.591	2.06	$\ln(AGE)$	2.987	2.49
US	-0.181	-1.52	%MALE	-0.017	-2.61
WTPAVOID	0.799	1.72	US	-0.101	-0.73
JOURNAL	-0.357	-2.05	WTPAVOID	0.516	1.25
CONSTANT	-12.031	-1.66	JOURNAL	-0.382	-1.96
			CONSTANT	-13.293	-2.02
R^2	63.50%			NA	
<i>N</i>	236			235	

^a Estimated with weighted least squares regression (SAMPLE SIZE weight).

^b Estimated with weighted maximum likelihood regression (SAMPLE SIZE weight).

^c Based on robust standard error estimates, corrected for clustering by study.

^d See Equation (9) for interpretation of coefficient.

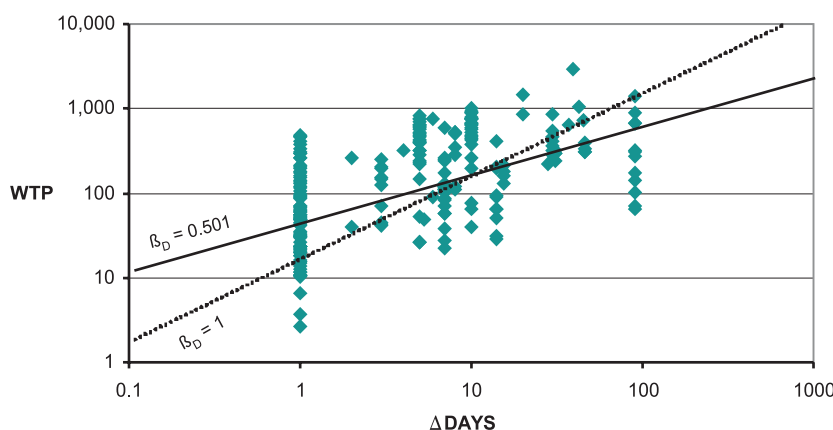


Figure 1. Observed and fitted WTP estimates with respect to duration of illness $^a\beta_D = \partial \ln(\text{WTP}) / \partial \ln(\Delta \text{DAYS})$

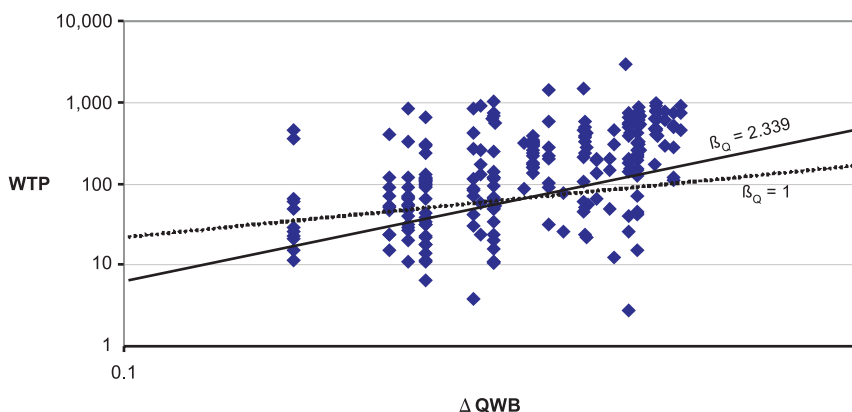


Figure 2. Observed and fitted WTP estimates with respect to severity of illness $^a\beta_Q = \partial \ln(\text{WTP}) / \partial \ln(\Delta \text{QWB})$

separate scores (last specification in Table 5). We selected these specifications because, as argued previously, they are most defensible from an *a priori* standpoint and because they provide strong empirical results in terms of goodness of fit and statistical significance.

In both cases, we conducted specification tests and dropped the least significant variables. The revised specifications are reported in Table 6 as BT Function 1 and BT Function 2, respectively. For BT Function 1, the three variables related to survey method (OPEN ENDED, PAYMENT CARD, and IN PERSON) and the gender proportion variable (%MALE) were dropped from the original specification. Based on an *F*-test, the coefficients for these four variables were jointly not significantly different from zero. For BT

Function 2, only the three survey method variables were dropped from the original specification. Using a likelihood ratio test, a restricted model that also held the %MALE coefficient at zero could not be rejected at a 0.05 level of significance.

Before evaluating the out-of-sample implications of these models, it is informative to compare the fitted model with the estimation sample. Figures 1 and 2 plot the 236 WTP estimates used to estimate BT Function 1 against their respective ΔDAYS and ΔQWB values. Although there is some clustering of ΔDAYS and ΔQWB on specific values, the plots show a reasonably good dispersion of data in both cases (on the logarithmic scale). There is also a loose but notable upward trend in WTP with respect to both ΔDAYS and ΔQWB . Both figures also include a fitted

regression line based on the parameter estimates for BT Function 1. The dark line in Figure 1 shows the model predicted value for $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{DAYS})$, holding all other explanatory variables at their respective means. For ease of interpretation, $\ln(\Delta\text{DAYS})$ and $\ln(\text{WTP})$ are shown without logarithmic transformation on their respective axes. A similar approach is used in Figure 2 for $\ln(\text{WTP})$ and $\ln(\Delta\text{QWB})$. Both of these lines can be compared to the dotted line, whose slope represents unit elasticity (i.e. the assumed elasticity under the QALY valuation approach). The fitted slope of $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{DAYS})$ is clearly less than unit elasticity, and the opposite is true for the fitted slope of $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{QWB})$.

To explore the implications of the BT functions under selected out-of-sample conditions, we applied each function to predict average individual WTP for four selected acute conditions. As shown in Table 7, each of the selected acute conditions corresponds to a different severity description, which is defined using the QWB health dimensions and scores. The symptom descriptions vary from relatively minor ('breathing unpleasant air') to relatively severe ('sick, vomiting'), and the three functional restrictions – mobility, physical activity, and social activity – are alternatively moderate or severe. For each of the four conditions, the duration of illness avoided (ΔDAYS) is also evaluated at either 1 or 10 days.

For both BT Function 1 and BT Function 2, Table 7 shows the mean predicted WTP and corresponding 95% confidence interval (CI) for each condition and duration (eight combinations). Each predicted value is estimated using the specified duration and QWB scores. The values of the other explanatory variables are held constant across the eight scenarios. In all cases, INCOME is set to \$45 000 per year, AGE is set at 45 years, and for BT Function 2 the %MALE is 50%. For these illustrations of the BT functions, the values are also all estimated assuming a US population ($\text{US} = 1$) and for avoiding health declines ($\text{WTP}_{\text{AVOID}} = 1$). Furthermore, assuming that values published in peer-reviewed articles are most defensible, we set JOURNAL equal to one as well. To appropriately transform the model prediction of mean $\ln(\text{WTP})$ to mean WTP in nonlogarithmic form, we applied a smearing factor to the model-predicted values [36]. The smearing factor is equal to the average of the exponentiated residuals for the observations in the estimation sample.

For BT Function 1, the predicted mean WTP values range from \$45 (95% CI: \$39–\$51) for avoiding one day, poor air and moderate functional restrictions to \$706 (95% CI: \$345–\$1067) for avoiding 10 days of sickness and vomiting with severe functional restrictions. For each condition, the 10-fold increase in duration has a positive but closer to 3-fold effect on predicted WTP. In contrast, increasing total ΔQWB by a factor of two (between conditions 1 and 2) increases predicted WTP substantially, by a factor of roughly 5.

For BT Function 2, the predicted mean WTP values show less variation. They range from \$124 (95% CI: \$68–\$181) to \$689 (95% CI: \$227–\$1150). As expected, based on the parameter estimates for this function, predicted WTP is somewhat less sensitive to differences in the duration of illness. Predicted WTP is also less sensitive to variations in severity using BT Function 2. For the least severe condition, predicted WTP is nearly three times as large as the WTP estimate based on BT Function 1; however, it is roughly the same of the most severe condition. In particular, BT Function 2 is relatively less sensitive to changes in the QWB symptom scores (holding the functional restriction scores constant, as between conditions 1 and 2 and between conditions 3 and 4).

For comparative purposes, Table 6 also includes value estimates based on a simple QALY valuation approach (using QWB as the health utility index) for each scenario. In five of the eight scenarios, these QALY-based estimates are higher than one or both of the BT function estimates. They are particularly large compared to the BT function estimates when the duration change is large (e.g. 10 days) and/or when the severity of illness is lower. These differences are consistent with the different slopes shown in Figures 1 and 2. They occur because the QALY-based approach assumes that values increase in direct proportion to duration and to severity, whereas the BT functions have a lower elasticity (< 1) with respect to duration and a larger elasticity (> 1) with respect to severity.

Conclusions

Given the increasing importance placed on CBA for evaluating public health and safety policies, policy makers need to be able to consistently